AUTONOMOUS DRONE MISSION

INTRODUCTION

The Autonomous Drone Mission is designed to test the ability of UAV systems to operate independently in structured environments. This competition focuses on developing drones that can perceive surroundings, map the locations autonomously, process data, and execute precise movements without manual control. It reflects real-world applications such as surveillance, inspection, and smart navigation, where reliability, accuracy, and autonomy are essential.

PROBLEM STATEMENT

Design and develop an autonomous drone capable of performing complete end-to-end navigation within a structured environment without any human intervention. The drone must autonomously take off, detect and interpret environmental cues, and navigate through a predefined or dynamically assigned path using onboard sensors, vision systems, or intelligent algorithms.

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The system should demonstrate stable flight performance, precise control, and reliable decision-making throughout the mission. It must be capable of identifying and avoiding obstacles, accurately reaching designated waypoints, and executing controlled hovering at specified checkpoints.

In addition, the drone must achieve high-precision landing at designated target locations, ensuring accurate positioning within defined landing zones as part of the mission objectives.

DRONE SPECIFICATIONS

- Recommended flight time: 20 minutes (not a strict condition; no marks deduction)
- Frame size: 300–550 frame, maximum 10-inch propellers allowed
- The drone must have a reliable transmitter and receiver system capable of operating in environments with walls and structures that may cause signal interference
- Battery voltage: ≤ 24V
- The drone must be capable of fully autonomous operation during the mission

ABSTRACT SUBMISSION ROUND

The participating teams must submit an abstract of their drone's concept, in proper documented form. This should include its configuration, design, features, estimated cost, and an approach to tackle the problem statement. The teams must compile all this information in a PDF file. If any extra files are to be included, the submission must be in the form of a ZIP file. Please note that the files submitted should NOT be password-protected or corrupted. In either of the cases, the submissions would be directly rejected.

GUIDELINES FOR ABSTRACT SUBMISSION

Rough sketches of your idea including the Design Report of your model. The abstract must specify the use of software and the mechanism used in the design analysis of the drone.

- USPs (Unique Selling Propositions) of your design and innovations made by you should be mentioned in the report and mention how they are going to tackle the specific challenges.
- An abstract of the idea needs to be uploaded on the Dronotics website before the deadline.
- The abstract should not exceed 500 words and should include the following details of all team members along with the Team.

Cover Page Details:

The cover page of the submission must include the following information:

- Team Name
- Team Leader Details:
  - Name of Team Leader
  - College / Institute Name
  - Year of Study
  - Email ID
  - Phone Number
- Team Members:
  - Names of all team members

File Naming Convention:

The file must be submitted in the following format:

ADM_<TeamName>_<LeaderName>_abstract.pdf

Example (for team ABC) : ADM_ABC_AmanGandotra_abstract.pdf

Submission Email

The abstract/design report must be mailed to:
dronoticsjiit128@gmail.com

The Subject of email should be in this format:
Autonomous Drone Mission - <TeamName> - <LeaderName> Abstract
Example: Autonomous Drone Mission - ABC - Aman Gandotra Abstract

Important Note

- Only PDF format submissions will be accepted.
- Files must not be corrupted or password-protected.
- Any submission not following the naming convention may be rejected.

ROUND 1

Round 1 is designed to evaluate the basic functionality, stability, and control performance of the drone under manual operation. This round serves as a preliminary screening stage to ensure that all participating drones meet the required operational standards before advancing to autonomous tasks.

- The round will be conducted in a structured arena of dimensions 15 m × 10 m.
- Participants will be required to manually pilot their drone throughout the arena.
- The drone must operate at altitude of approximately 6 meters during the entire flight.

GUIDELINES FOR ROUND 1

This round will be conducted in manual mode only; autonomous features will not be considered. Participants must ensure stable flight, smooth maneuvering, and full control throughout the task

Pre-Round Inspection

- Before the start of Round 1, an inspection round will be conducted for each team
- The drone will be checked to ensure compliance with all design and specification constraints
- In case of any deviation, penalties will be imposed on the overall score (Round 1 + Round 2)

Penalties

- Any increase beyond the allowed frame size will result in a 10% penalty
- For every 2-minute increment taken to complete the mapping task, a 5% penalty will be applied to Round 1 marks

ROUND 2

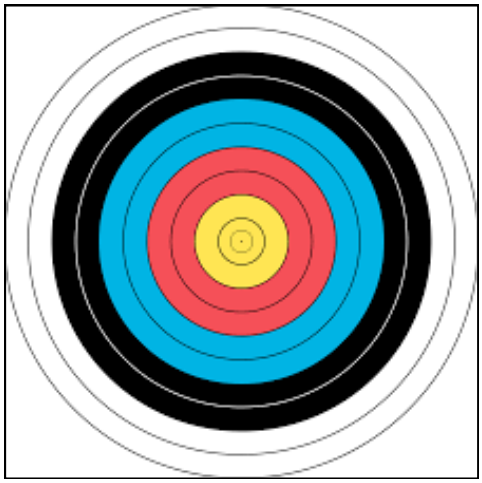
Round 2 evaluates the autonomous capabilities, decision-making, and precision performance of the drone in a real-world inspired mission scenario. The drone must complete the entire task without any manual intervention after takeoff.

- Each team will be provided with two flight attempts
- The mission must be executed fully autonomously once initiated

The mission consists of two major phases:

1. Mapping & Target Detection

- The drone must autonomously explore and map the designated area
- There will be 3 Target Locations
- It must detect and identify target locations based on reference images provided prior to the run
- The appearance of targets (such as color shades) may vary during the event
- The system should be robust enough to accurately detect targets despite such variations



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2. Autonomous Landing Mission

- After detecting the targets, the drone must:
  - Navigate to three designated landing locations sequentially
  - Perform a controlled landing at each location
  - Hold position (stay landed) for at least 3 seconds at each point
  - Take off again and proceed to the next location
- The landing surfaces:
  - May be uneven or inclined
  - Exact conditions will be disclosed on the day of the event
- After completing all three landings:
  - The drone must take off again and return to the original takeoff position
  - Perform a final landing at the starting point

SCORING CRITERIA

Round 1: Judgement Round (100 Marks)

- Successful mapping: 20 marks
- Detection of waypoints: 70 marks
- Bonus (if applicable during the event): 10 marks

Round 2: Autonomous Mission (100 Marks)

- 90 marks will be distributed equally based on successfully reaching each waypoint (landing targets)
- 10 marks will be allocated as bonus marks

Bonus Marks (Applicable in Both Rounds) may be awarded for, but are not limited to:

- Implementation of fail-safe mechanisms (e.g., safe landing during signal loss or low battery)
- Controlled and precise landing in emergency or low-power conditions
- Efficient battery management and overall system reliability

Note: The above are indicative criteria. Additional bonus opportunities may be announced during the event at the discretion of the organizers.

Note: The organizers reserve all rights to change any or all of the above rules.